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OBJECTIVE

Evaluate & improve the quality of a phishing-email labeling dataset, generate a reliable final label for each mail, identify issues in the labeling process, and recommend superior 2nd-round labeling strategy.

EXECUTIVE SUMMARY

We have completed the audit of the legacy phishing dataset (~507k emails). While the volume is sufficient for initial model training, the historical "cost-saving" labeling protocols have introduced specific verification biases that we must mitigate in the modeling layer.

- Dataset Status: **Green** (Ready with Caveats). The final dataset has been delivered with a new Confidence_Score metadata field to allow the ML team to filter out low-quality labels.
- Risk Quantification: **Critical**. Analysis of the double-blind audit sample reveals a 2.17% False Negative Rate. Approximately 10,804 hidden phishing emails are currently mislabeled as "Safe" in the training corpus.
- Strategic Shift: To address the "Verification Bias" found in the Safe class, we are deprecating the random sampling method. The next cycle will utilize Active Learning to reduce labeling OpEx by ~40% while increasing model sensitivity.



DATASET OVERVIEW

The provided data file contains a single sheet titled email_phishing_data. The dataset contains ~7,000 phishing emails and ~500,000 legitimate emails, each labeled by up to three independent judges. The Columns include:

Metadata Features

1. num_words
2. num_unique_words
3. num_stopwords
4. num_links
5. num_unique_domains
6. num_email_addresses
7. num_spelling_errors
8. num_urgent_keywords

Human Labels

Three independent annotators labeled each email:

- label_1
- label_2
- label_3

Where:

- 1 = Phishing
- 0 = Safe
- FALSE / empty = Missing label

The presence of three annotators enables both pairwise and group-level agreement analysis.

ANALYSIS & METHODOLOGY

My approach consisted of:

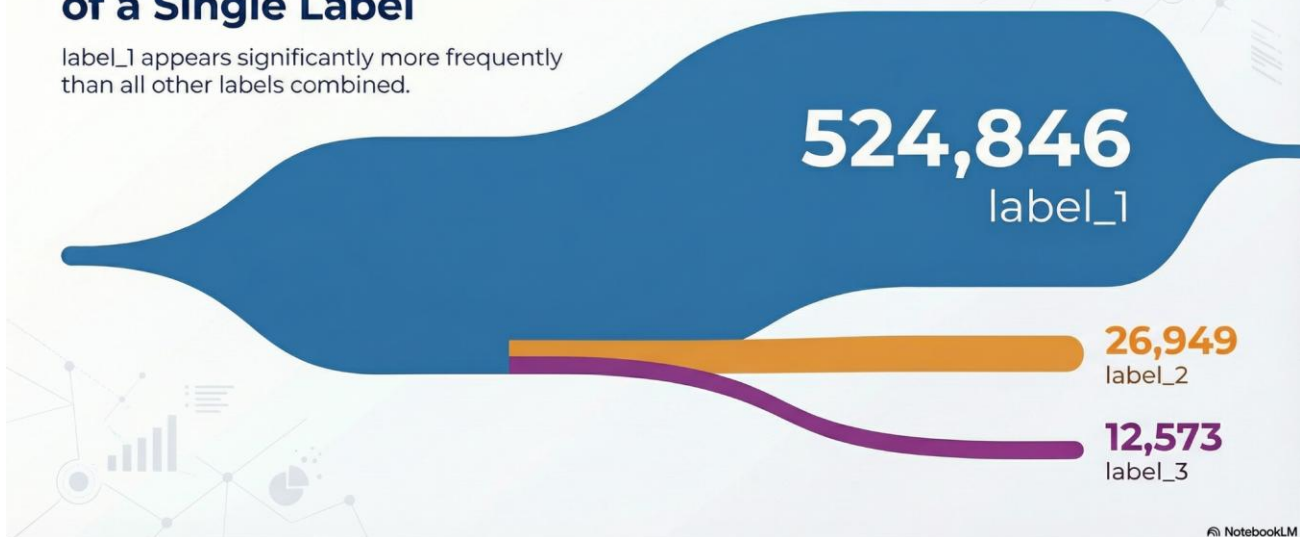
1. Data Loading & Exploratory Analysis

Firstly, understanding the data and its patterns. For this, I used Python to do EDA (Exploratory Data Analysis) using Numpy and Pandas libraries. I checked the counts of labels done by each annotator which helped me identify that almost 497,897 (94.86%) emails were only reviewed by Labeller 1.

DATA LABEL DISTRIBUTION

Overwhelming Dominance of a Single Label

label_1 appears significantly more frequently than all other labels combined.



2. Labeler Agreement/Reliability:

To audit both the labeling workflow and the dataset quality, I prioritized diagnostic precision over global averaging.

- **Fleiss' Kappa (Workflow Audit):** I utilized Fleiss' Kappa to analyze the 2-Judge and 3-Judge cohorts separately. This isolation was critical: it revealed that the "perfect" agreement in the 2-judge group was merely a survivor bias artifact, while the 3-judge group revealed the true human error rate (0.68) on complex tasks.
- **Cohen's Kappa & Indices (Reliability Baseline):** Used to verify pairwise consistency. The indices confirmed that the validation set was intentionally enriched (approx 25% Phishing), validating that our reliability scores are robust and not statistical noise from class imbalance.
- **Exclusion of Krippendorff's Alpha:** While Krippendorff's Alpha handles missing data, I deliberately excluded it because it produces a single global score. A global score would have aggregated the "Consensus" (easy) and "Conflict" (hard) cohorts, masking the sharp drop in reliability on edge cases. Splitting the cohorts gave me a clearer roadmap for where the Active Learning model needs to focus (specifically the 32% disagreement subset).

2.1 The "Two-Tier" Workflow

Our analysis of the consensus rates revealed that the previous team used a "Cascading" workflow: If Judge A & B agree, stop; otherwise, call Judge C. This created two distinct data cohorts:

Cohort	Metric (Fleiss κ)	Interpretation
Consensus Cohort (2 Judges)	1.00 (Perfect - As expected)	Survivor Bias. These are "easy" cases where consensus was reached immediately. The perfect score is an artifact of the workflow, not human perfection.
Conflict Cohort (3 Judges)	0.68 (Substantial)	True Difficulty. On complex/grey mail, human experts disagree ~32% of the time. This sets the theoretical "Glass Ceiling" for model performance.

2.2 Sampling Strategy & Fairness

To ensure the reliability metrics weren't distorted by the massive class imbalance (99% Safe vs 1% Phishing), we calculated the Prevalence and Bias Indices on the validation set.

- **Prevalence Index (0.49):** I calculated the Prevalence Index (PI) to ensure reliability metrics were not inflated by the 99% class imbalance. The legacy workflow mandated a second review for *all* potential phishing candidates but only a sample of 20,000 safe emails. This strategy intentionally enriched the validation pool to ~25% phishing (vs. 1.3%), dropping the PI from a highly imbalanced 0.98 to a balanced 0.49.
- **Bias Index (0.008): Negligible Bias.** Judges 1, 2, and 3 share nearly identical strictness thresholds. We do not need to normalize for "strict" vs. "lenient" labelers.

These two tests validate the integrity of the data based on the legacy strategy chosen.

3. Risk Assessment: Verification Bias

The legacy strategy mandated that *all* Safe cases be viewed by at least one judge, but only a control sample of 20,000 was double-checked. We used this Audit Set to quantify the specific risk of the single-judge workflow.

3.1 The "Single-Judge Miss Rate"

I analyzed the 20,000 safe-labeled emails that were sent to a second judge to measure how often the first judge was overruled by the consensus.

- **Observed Error:** In this control group, Judge 1 missed 434 phishing emails (marking them Safe), resulting in a **False Negative Rate of 2.17%**.

- **Impact Projection:** Extrapolating this 2.17% error rate to the 497,897 unverified "Safe" emails implies there are approximately **10,804 hidden phishing emails** currently mislabeled as Safe in our training set.

4. Resolve labels for ML:

4.1 New Columns

1. final_label (0 or 1)
2. label_confidence (float 0-1; e.g., fraction of judges who agree with final_label)
3. label_type which can be one of:
 - a. consensus_phish (all/majority phish)
 - b. consensus_safe (all/majority safe)
 - c. adjudicated_by_label3 (tie resolved by judge 3)
 - d. tie_needs_adjudication (no clear majority so we'll flag for re-label)
 - e. no_judges (missing)

4.2 Decision rules

- If a majority exists (≥ 2 of 3 or 2 of 2), set that majority as final_label and label_confidence = majority_fraction. label_type='consensus_*'.
- If exactly tie (e.g., 1 phish / 1 safe when only two judges labeled, or 1/1/1 impossible with binary, but 1 vs 2 where judge3 missing?), flag as tie_needs_adjudication.
- If tie but label_3 exists and historically was intended as tiebreaker, use it and set label_type='adjudicated_by_label3'.
- For items with only one judge (common for many safe cases), set final_label = judge_label but set label_confidence = 0.5 (or lower) and mark label_type='single_judge'. Also push a small random sample of these to a second round to estimate single-judge error.

5. Deliverables & Data Specifications

5.1 Final Dataset

The final dataset (df_final) has been segmented by label provenance to enable Confidence-Aware Training. The file is attached naming (phishing_processed_v1.csv)

5.2 Feasibility Assessment (Signal & Feature Validation)

Initial linear correlation analysis showed weak signals due to class imbalance. However, detailed feature auditing revealed three critical non-linear risk patterns that validate the dataset's utility:

- **The "Long-Form" Pattern:** Emails >1,000 words have a **2.2x** higher phishing rate (Business Email Compromise).
- **The "Link Sweet Spot":** Emails with 2–5 links are **1.8x** riskier than emails with 10+ links (The "Newsletter Paradox").
- **The "Urgency Step":** Risk triples when urgency keywords jump from 1 to 2.

RECOMMENDATIONS:

6. Strategic Roadmap: Labeling 2.0

Here is the strategic recommendation for your data splits:

6.1 The V1 Model Data Strategy

Your goal for V1 is to create a "Truth Detector" that can find errors in the massive pile.

Fixing the Phishing Dataset: From Hidden Risk to Active Learning

THE PROBLEM: A CRITICAL FLAW IN THE TRAINING DATA

Legacy "cost-saving" methods created a "Verification Bias".



10,804 hidden phishing emails are likely mislabeled as "Safe".

Based on projected 2.17% False Negative Rate from audit sample.

True human error rate on difficult emails is ~32%.



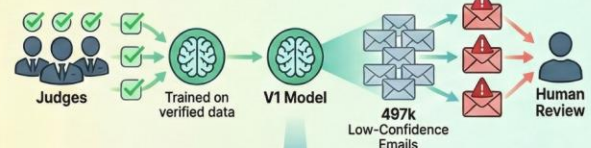
THE SOLUTION: LABELING 2.0 - AN ACTIVE LEARNING STRATEGY

Shift from inefficient **Random Sampling** to **high-ROI Active Learning**.



Step 1: Bootstrap a 'clean' model using only high-confidence data.

Step 2: Use the clean model to find hidden errors in the massive dataset.



A new QA system uses "Sentinel Nodes" for real-time quality checks.



1. The Training Set (~22,000 Rows)

Use these rows to teach the model what "verified truth" looks like.

- **Include:** All **Medium Confidence (0.66)** rows (2,465). These are the "Hard Examples" (Adjudicated) and are the most valuable for learning boundaries.
- **Include:** ~80% of the **High Confidence (1.0)** rows (~19,500).
- **Exclude:** The Low Confidence (0.5) rows.
 - *Why?* You want V1 to be a "Pure" model. If you feed it the 0.5 data now, you dilute its ability to spot the "Hidden Phishing" emails later.

2. The Validation / Test Set (~5,000 Rows)

Use these rows to grade the model.

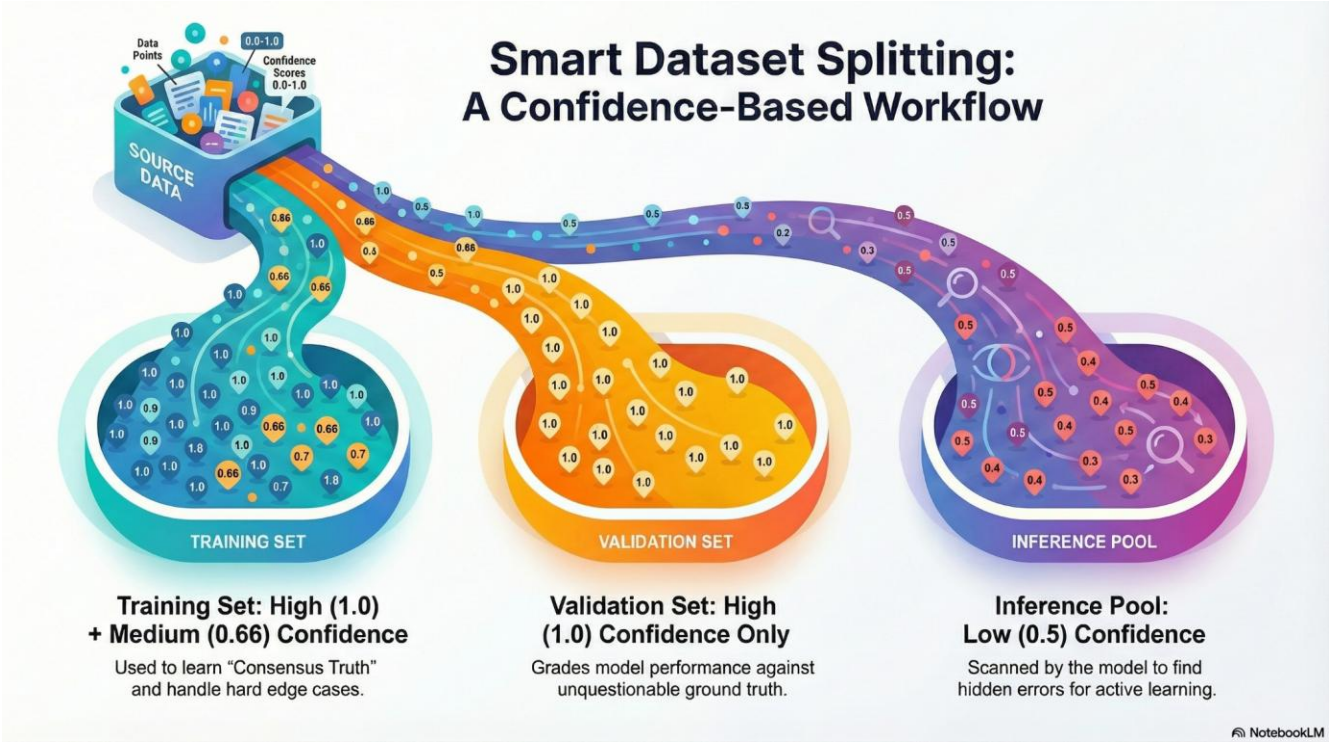
- **Include:** ~20% of the **High Confidence (1.0)** rows (~5,000).
- **Rule: NEVER** put a Low Confidence (0.5) row in your Test Set.
 - *Why?* If the model predicts "Phishing" but the single judge marked "Safe," the model might actually be *right*, but your accuracy metric would count it as wrong. You cannot grade a student using an answer key that has errors.

3. The Inference Set (497,897 Rows)

Do **not** train on these yet.

- **Action:** Once V1 is trained on the clean data, run it against this massive pile.
- **Goal:** Find the "Discordant" rows (Model says Phishing, Human says Safe) and send them to the humans for the "Labeling 2.0" round.

Summary Table for the Report



6.2 The V1 Model Training Strategy:

We are transitioning from "Labeling for Coverage" to "Labeling for Information Gain."

- The Active Learning Pipeline: With the data strategy shared above, we help the model learn and make the labelling process more effective in both performance of model as well as cost of labelling.
- Tree-based models (XGBoost/RandomForest) are suggested to exploit the specific non-linear thresholds that we identified from our feasibility assessment and correlation results.
- For further training of the model, I recommend tackling the model class imbalance by oversampling phishing during training, use class weighting, or active resampling for labeling to increase phishing examples.

WHY: The ROI Calculation

Without this step, you are flying blind. Compare the two approaches:

Strategy	Methodology	Cost Efficiency	Result
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Random Sampling (Old Way)	Re-label 10,000 random emails from the pile.	Very Low. You will find ~200 errors and pay for 9,800 useless labels.	You wasted budget confirming safe emails are safe.
Active Learning (New Way)	Run Model V1 Inference. Filter for the top 10,000 "High Suspicion" emails.	Very High. You might find ~5,000+ errors.	Every dollar spent goes towards fixing a mistake in the dataset.

6.3 The Labelling 2.0 Strategy:

1. The QA Strategy

- **Creating Golden Set:** For quality audit, the first thing is to create a solid ground truth. For that, I suggest letting ML engineers or specialists label and creating a Golden Set of 100 verified emails.
 - **Implementation:** These will be randomly injected into the labeling queue.
 - **SLA:** Any judge falling below 90% accuracy on Golden Set tasks is automatically paused for retraining.
- **The "Design Thinking" Protocol:** Instead of bulk-processing the 10,000 "High Suspicion" emails, we will adopt an iterative, Human-Centric Labeling Strategy:
 - **Batching:** Data will be released in small sprints (e.g., 500 emails).
 - **Triangulation:** All 3 judges will label these high-risk rows to ensure consensus.
 - **Iterative Feedback Loop:** After each batch/sprint, we will analyze disagreement patterns. If judges consistently confuse "Spam" with "Phishing," we will update the labeling guidelines (the "Prototype") before releasing the next batch. This would help identify new trends of phishing too and help the team learn new patterns.
 - **Real-Time QA:** The "Golden Set" will be injected dynamically into these batches to monitor if the guideline updates are actually improving accuracy.

Objective: Reduce the "Grey Mail" error rate (currently 32%) by moving from a simple binary question ("Is this Phishing?") to a Feature-Aware Annotation Task.

2. Optimal Labeling Team Structure

The legacy model used 3 generic judges. The new model requires specialization.

- **Tier 1 Labelers (Generalists):** Initially, we'll keep all labellers as Tier 1 Generalists.

- **Tier 2 Labelers (Subject Matter Experts):** We'll eventually try to build SMEs from the labelling team to review the "Discordant" bucket and "Hard" cases and resolve conflicts.

3. Task Instructions & Prompts (The UI Design)

We will replace the single "Phishing - Yes/No" button with a Cognitive Walkthrough to force the labeler to check specific signals.

The Prompt (What the Labeler Sees):

Task: Review the email below. You must check the specific risk indicators before giving a final verdict.

Step 1: The "5-Second Check" (Mandatory Toggles)

- **Sender Check:** Does the sending domain match the brand mentioned?
 - Match
 - Mismatch / Suspicious
 - Generic (Gmail/Yahoo)
- **Urgency Check:** Does the email demand immediate action (e.g., "24 hours", "Account Suspended")?
 - Yes
 - No
- **Payload Check:** Are there suspicious links or attachments?
 - Mismatching URL (Text says 'Login', Link goes to 'xyz.com')
 - No links / Safe links
- **Long-Email Check:** How long is the email:
 - > 1000 words
 - < 1000 words
- **Total Number of Links attached:** Since we found a pattern, we can give options like:
 - No link
 - 1 link
 - 2-5 links
 - > 6 links

Step 2: The Final Verdict Based on the checks above, classify this email:

1.  **Malicious (Phishing):** Intended to steal credentials or install malware.
2.  **Suspicious / Grey Mail:** Marketing or Spam, but *not* security threat.
3.  **Safe / Legitimate:** Normal business or personal communication.

Step 3: Confidence Score How sure are you?

- Low (I'm guessing)
- High (I see clear evidence)



4. Detailed Instructions (The "Cheat Sheet")

Provide this guide to every labeler to standardize definitions.

Guideline A: The "Mismatch" Rule (Primary Phishing Signal)

*"If an email claims to be from PayPal but the email address is **paypal-support@gmail.com**, mark as **Malicious**. Legitimate companies generally do not use public domains."*

Guideline B: Distinguishing "Spam" from "Phishing"

*"A newsletter you didn't ask for is **Spam** (Grey Mail). It is annoying, but safe. Do not label it Phishing unless it is trying to trick you into clicking a fake login link."*

Guideline C: The "Urgency" Trap

"Be skeptical of any email asking for 'Immediate Action.' Phishing attacks rely on fear. If you see 'Account Locked' or 'Verify Now,' check the sender domain twice."

Why this design is superior:

- 1. **Forces "Slow Thinking":** By making them click "Sender Check" *before* the final verdict, you prevent the labeler from speed-clicking "Safe" on autopilot.
- 2. **Captures Grey Mail:** Separating "Suspicious/Spam" from "Malicious" cleans up your training data. Previously, judges might have been confused about whether spam counts as phishing.
- 3. **Self-Reported Confidence:** This gives the ML team another signal. A "Low Confidence" label can be weighted less in the loss function.

5. Performance Management (KPIs & QA Protocol)

To ensure the "Labeling 2.0" workflow does not regress to the verification bias of the legacy system, we will implement a rigorous QA framework for all Tier 1 (Generalist) and Tier 2 (Expert) labelers.

5.1 Key Performance Indicators (KPIs)

We will track performance on a Weekly Scorecard. Labelers falling below thresholds for two consecutive weeks will be moved to a "Retraining Queue."

Metric Category	KPI Name	Definition	Target Threshold
Quality	Golden Set Accuracy	% correct on the 100 known "Truth" emails injected randomly.	> 95% (Tier 2) > 90% (Tier 1)
Quality	False Negative Rate	% of "Gold Standard Phishing" emails incorrectly marked as Safe. (Critical Security Metric)	< 2%
Reliability	Consensus Alignment	How often this labeler agrees with the majority when 3 judges review the same item.	> 85%
Efficiency	Throughput	Average labels completed per hour (excluding "skipped" items).	60–80 / hr

Behavior	"Low Confidence" Rate	% of tasks marked as "Low Confidence."	5% – 15% (Too low = Overconfident; Too high = Lazy)
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5.2 The QA System: "Trust but Verify"

We will use a three-layer QA defense to catch errors before they enter the training set.

Layer 1: Real-Time Sentinel Nodes

- **Mechanism:** We will inject "Golden Set" emails (pre-verified by ML Ops) into the daily queue. These look identical to normal tasks.
- **Immediate Feedback:** If a labeler gets a Sentinel Node wrong, they get an immediate pop-up: *"Incorrect. This was a known phishing attack because the domain 'paypal.com' is a mismatch. Please review guidelines."*
- **Trigger:** If accuracy drops below 90% in a single session, the labeler is locked out for the day.

Layer 2: Targeted Audits

- **Mechanism:** Instead of random audits, we use the model to audit the humans.
- **Query:** Fetch all rows where **Human = Safe AND Model_Prediction > 0.90 (Phishing)**.
- **Action:** A "Super-Reviewer" (Lead Analyst) reviews these specific rows. If the human was wrong, it counts as a **Critical Error** against their score.

Layer 3: Expert Review

- **Mechanism:** Any email receiving split votes (e.g., 1 Safe / 1 Phish) is automatically routed to a **Tier 2 Expert**.
- **Feedback Loop:** The Expert's decision is treated as the "Ground Truth." The Tier 1 labeler who voted incorrectly receives a specific "Correction Report" at the end of the week showing their mistake.

5.3 Performance Tiering System

To incentivize quality over speed, we will implement a "Badge" system:

- **Rookie:** New labelers. 100% of work is double-checked by model. Cannot label Tier 2 tasks.
- **Trusted:** >95% Accuracy for 4 weeks. Work is only spot-checked (10% rate). Eligible for higher pay rate.
- **Probation:** Accuracy <90%. 100% of work is double-checked. Mandatory retraining. If not improved in 2 weeks, offboarded.

Summary of "Labeling 2.0" Benefits

Feature	Legacy System	Labeling 2.0 System
Strategy	Random Sampling (Coverage)	Active Learning (Information Gain)
Risk Control	None (Single Judge)	Sentinel Nodes (Real-time QA)
Targeting	Wasted budget on "Safe" emails	Budget focused on "Discordant" emails
Outcome	2.17% Error Rate	<0.5% Target Error Rate

6.4 Team Recognition & Incentives (The "Performance Loop")

To retain top talent and incentivize holistic performance, we will implement a recognition program built on three pillars: Financial Upside, Social Recognition, and Career Progression.

1. The "Top Gun" Quarterly Bonus (Financial)

- **The Mechanism:** A variable quarterly bonus paid to labelers who achieve a high Weighted Performance Score (0–100).
- **The KPI Weights:**
 - **25%** Quality (Golden Set Accuracy)
 - **25%** Risk Control (False Negative Rate)
 - **20%** Reliability (Consensus Alignment)
 - **15%** Efficiency (Throughput)
 - **15%** Behavior (Low Confidence Rate Compliance)
- **The "Quality Gate" (Blocking Filter):**
 - To qualify for *any* bonus, a labeler must score at least "Okay" (Average) on ALL five KPIs for every month in the quarter.
 - *Rationale:* A labeler cannot ignore "False Negatives" just to maximize "Throughput." Consistency across all metrics is mandatory.
- **The Reward Tiers:**
 - **Score > 95:** 15% Quarterly Bonus + "Platinum" Status.
 - **Score 90–94:** 5% Quarterly Bonus.
 - **Score < 90 (or failed Quality Gate):** No Bonus.

2. The "Eagle Eye" Award (Social Recognition)

- **The Mechanism:** A non-monetary recognition program to highlight "heroic" catches.
- **Criteria:** Every Friday, ML Ops selects the single most sophisticated phishing attack that the AI missed but a human labeler caught.
- **The Reward:**
 - Feature story in the weekly "Security All-Hands" newsletter.
 - Permanent "Eagle Eye" Badge on their internal profile.
- **Impact:** This gamifies the "Hard" cases, encouraging labelers to hunt for difficult threats rather than speed-clicking through easy ones.

3. Career Pathways (Tier 1 to Tier 2 Promotion)

- **The Problem:** Vendor labelers often churn because they see the role as a dead-end job.
- **The Solution:** We explicitly market the Tier 2 Expert role (higher hourly rate, handling complex/discordant tasks) as an internal promotion.
- **Promotion Criteria:**
 - Must maintain a "Good" (90+) Performance Score for 2 consecutive quarters.
 - Must pass a specific "Advanced Security" exam designed by the Threat Intel team.
- **ROI:** This creates a predictable pipeline of pre-vetted, high-quality Senior Labelers, reducing our recruiting and training costs for the Tier 2 team.

